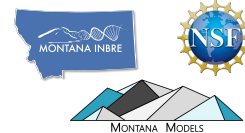




Modeling Wildfire Spread In Montana

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Abstract

Wildfires have become such a rising concern recently and their spread has a lasting impact globally. Wildfires spread at an average of 14.27 miles per hour, however, this can change and vary depending on different factors: weather conditions, fuel, types, terrain, etc. Montana is not new to wildfire. Before colonization, Indigenous people would purposely start fires to clear away shrubs and rubbish to make better grazing areas for livestock, protect themselves against enemy attacks, make paths, etc. As climate change continues to become a growing concern the number of wildfires has greatly increased and become more common and impactful. Using the program NETlogo, we made a simple model of how wildfire spreads throughout Montana. Through this model, we were able to see ways to minimize the spread and learn more about ways to protect ourselves.

Objective

My objective was to create an activity for youths that were participating in a summer camp known as **Montana Models**.

Montana Models brings rural youths from across the Montana state to Montana State University (MSU) campus for a summer camp focusing on mathematics and mathematical modeling.

What is Mathematical Modeling?

Mathematical models are representations of the world that people use to understand phenomena and make decisions.

In mathematical modeling, we ask questions that point us towards mathematics.

NetLogo

Netlogo is a software that allows users to model the environment. Using Netlogo, we designed a model of Montana so that the students get a chance to see how wildfire spreads throughout their state. The program is extremely user friendly and very hands-on. There were many different features included such as **forest density**, **mountain ranges**, **river**, and **wind**. All these features can be controlled and give a representation of how wildfire spreads and the different factors that increase/decrease the spread.

Background - Wildfire

Wildfires start with the help of three key elements, commonly referred to as the “fire triangle,” heat, fuel, and oxygen.

- **Heat** - helps to ignite the fire and causes the fire to spread
 - Sources of heat can come from either a natural source or human causes. (Most wildfires are caused by careless human activities)
- **Fuel** - anything that feeds a fire
 - Higher temperatures (caused due to climate change) creates more fuel for wildfires
- **Oxygen** - completely impossible to control

